AVLEE SINGH

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EDUCATION

Vellore Institute of Technology (VIT) Aug 2023 – May 2027
B.Tech in Mechanical Engineering *Vellore, India*

EXPERIENCE

Defence Research & Development Organisation (DRDO) May 2025 – Jun 2025
Research Analyst Intern *Hyderabad, India*

- **Accelerated scientific decision-making timelines by 44%** by synthesizing gigabytes of raw structural sensor telemetry into structured analytical findings for joint Indian and Russian defense scientist reviews.
- **Engineered high-velocity tracking workflows under 3x accelerated timelines** for high-stakes missile fuselage evaluations during high-priority operational deployment windows.
- **Ensured 100% adherence to defense safety protocols** across multi-category stress testing (load, vibration, fatigue) while maintaining discreet data coordination workflows.

Key Skills: Executive Data Reporting • Telemetry Analytics • Cross-Functional Alignment • Stakeholder Management

LEADERSHIP

BETI AI: Voice-First Digital Assistant for the Elderly Ongoing
Founder & Product Lead *India*

- **Defined 0-to-1 product strategy and core feature specs** for an elderly accessibility app by synthesizing qualitative pain points from **20+** user discovery interviews (ages **60+**) into a prioritized PRD roadmap.
- **Architected a voice-first UI wireframe layout** and low-latency interaction flow; integrated Sarvam API for natural Indian dialect TTS and Gemini API for LLM knowledge task completion.
- **Spearheaded product validation and pitch execution** for the Youth Entrepreneurs Challenge, executing continuous feature iteration cycles ahead of beta deployment.

Key Skills: 0-to-1 Product Strategy • User Research • Wireframing (Figma) • API Integration • UX

PROJECTS

DevWrap: Hybrid Developer Telemetry & Analytics Tool Sep 2026

- **Identified community market gap for dev tool analytics**; conceptualized DevWrap desktop companion using **<15 MB** RAM/ROM background footprint to monitor active vs. idle tool ROI across Chrome, Cursor, VS Code, and Terminal.
- **Acquired 55+ active users in 3 weeks** and executed **6 product iterations** based on direct user feedback telemetry and session heartbeat analytics.
- **Engineered a monthly "Dev Wrapped" viral loop** inspired by Spotify Wrapped and Apple Music Replay, aggregating tool usage into shareable social infographics to boost organic user acquisition.

Key Skills: Product Analytics • Growth Loops • Iterative User Feedback

Real-Time Drowsiness Detection & Alert Safety System Oct 2025 – Dec 2025

- **Won 1st Prize at Safety Hackathon** by identifying real-world driver fatigue risks from personal user discovery (long-distance driving safety) and building an end-to-end edge computer vision prototype.
- **Achieved sub-100ms detection latency** by engineering a Python, OpenCV, and Dlib pipeline tracking facial landmarks to calculate real-time Eye Aspect Ratio (EAR) drops.

Key Skills: Product Prototyping • Computer Vision (OpenCV) • Edge Deployment • Rapid Iteration

Alloy Property & Stress-Strain Predictive Pipeline Apr 2025

- **Accelerated materials screening workflows by 80%** by deploying a multi-output ML regression pipeline in Python that replaces expensive destructive tensile testing with predictive estimations (yield strength, UTS, Young's modulus) at **> 92%** accuracy.
- **Reduced prediction error (RMSE) by 25%** through systematic benchmarking of Random Forest, XGBoost, and Neural Networks combined with domain-specific feature engineering.

Key Skills: Machine Learning (Scikit-Learn/XGBoost) • Predictive Modeling • Workflow Optimization

RELEVANT SKILLS & CAPABILITIES

Product:	Product Strategy, PRDs & Specs, User Research, User Interviews, Feature Prioritization, Wireframing, Roadmap Planning, Agile/Scrum, Trello
Data & Analytics:	SQL, Python (Pandas, NumPy), Product Analytics, Excel Data Modeling, Data Visualization, A/B Testing
Technical & Tools:	C/C++, REST APIs, OpenCV, Scikit-learn, XGBoost, Git, Figma, Android Studio, Postman